QGen: Dynamic AI Question Generation During Live Classroom Lectures

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Abstract

University students occasionally struggle to ask questions in class. While AI can support learning, little is known about how it can support in-class question asking. We developed an AI system that generates questions in real time from lecture content and a mobile interface that allows students to view, vote on, and submit their own questions alongside AI-generated ones. We deployed this system in four computer-science lectures, and found that AI-generated questions helped students consider lecture content from different perspectives. For some, this prompted deeper reflection on what to ask. However, others reported reduced motivation to form their own questions, suggesting risks of overreliance. Although students found the AI helpful, they more frequently voted for questions asked by their peers, revealing a gap between AI-generated questions and students' in-the-moment learning needs. We provide design recommendations for systems that promote interactive collaboration with AI and scaffold question asking without undermining student agency.

CCS Concepts

• **Human-centered computing** → **Natural language interfaces**; **Empirical studies in HCI**.

Keywords

classrooms, educational technology, agents, language models, human-AI interaction, human-agent interaction

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1 Introduction

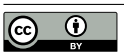
University students sometimes struggle to ask questions in class because they lack communication confidence or are unsure what to ask. For some, this difficulty reflects communication apprehension: even low-stakes participation can feel socially risky, leading students to avoid speaking publicly [33]. In multilingual classrooms, foreign-language anxiety can further increase the perceived cost of "speaking up," as students worry about making mistakes or being misunderstood [15]. In East Asian and Confucian-heritage contexts, norms of deference, harmony, and face can make public questioning feel particularly delicate; classroom "silence" may also be interpreted as respect rather than disengagement [5, 8].

Yet student questions are not optional. Formulating and articulating questions helps learners surface gaps in understanding, refine mental models, and support conceptual change, while also giving instructors a direct window into student thinking [7, 13]. More broadly, research suggests that shifting students from passive reception to more active and interactive engagement improves learning outcomes [6], and active-learning approaches outperform traditional lecturing across STEM courses [12].

The challenge is that many university lectures—especially large ones—offer limited time, psychological safety, or scaffolding for students to transform fleeting confusion into an askable question, particularly under rapid, slide-driven pacing [2, 11]. This motivates real-time, lightweight supports that help students generate, refine, and share questions without forcing them into the spotlight. Earlier automatic question generation (AQG) systems largely treated questions as products of offline NLP pipelines. A common strategy was to select informative sentences, parse them, and apply hand-built syntactic transformations to produce questions [21]. Practice-focused AQG often generated multiple-choice items by blanking key terms (e.g., named entities) and proposing distractors using lexical resources or similarity measures [35]. AQG also produced trigger questions tied to detected key concepts to prompt reflection [28]. However, these approaches assume clean and static input text, so they can miss students' immediate, context-sensitive curiosities and provide limited support for iterative question co-creation in live lectures.

Recent works bring AQG into classroom time by using LLMs to draft candidate questions directly from transient lecture traces. One recent empirical study uses time-sliced context: instructors segment instruction into short segments, and an LLM generates low-stakes, retrieval-practice multiple-choice questions with distractors. Instructors then review the items for issues such as multiple correct answers or implausible distractors [27]. Another line of work shows how LLMs can support real-time personalized notetaking from learners' micronotes [16]. Simulation-based work conditions generation on lecture transcripts while specifying structured student personas, producing diverse question styles [3]. Another

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AI-driven system uses a robot to visualize question-asking activity. However, the AI asks questions only after receiving lecture summaries instead of full live transcripts, and students did not speak up even after the robot and system were introduced [19].

We frame in-class question asking as a metacognitive, generative, and socially situated learning practice. Metacognitively, asking a question requires learners to monitor understanding, identify gaps, and express uncertainty in an actionable form [10, 44]. As scaffolding, AI-generated questions can make inquiry directions visible, but this support should be calibrated and gradually transferred back to students rather than replacing their own question formulation [41, 42]. From the perspective of the ICAP (Interactive, Constructive, Active, and Passive) framework, reading AI-generated questions may support active engagement, whereas revising, extending, or debating them can promote constructive and interactive engagement [6]. This framing motivates our design problem: how can AI-generated questions scaffold inquiry during live lectures while preserving students' agency to formulate, prioritize, and pursue their learning needs?

We developed QGen, an AI system that generates real-time questions from lecture content and displays them alongside student-submitted questions. Through a mobile interface, students can view, submit, and vote on both sources. Across four computer-science lectures, AI-generated questions encouraged reflection and exposed students to different perspectives, sometimes helping them formulate deeper questions. Students preferred peer questions, which better reflected their immediate learning needs, and wanted greater direct interaction with the AI. We therefore offer design recommendations for collaborative scaffolding that supports question asking while preserving student agency.

2 Related Work

Research on automatic question generation (AQG) has progressively widened its definition of a "good" educational question. Early systems treated the question primarily as a text artifact to be produced and evaluated. Heilman and Smith [14], for example, used an overgenerate-and-rank pipeline to filter questions with grammatical, semantic, or answerability problems. Lindberg et al. [26] combined semantic-role templates with educator judgments of acceptability, source relatedness, and usefulness, bringing pedagogical fit into the evaluation of generated questions. More recent work has sought finer control over cognitive level [40], topic [25], and contextual grounding through in-context learning and retrieval augmentation [32]. A complementary line of research models the questions students themselves might ask. Ros et al. [38] paired MOOC transcript windows with contemporaneous student questions and formulated answer-agnostic generation, showing that student inquiry often extends beyond information recoverable from the immediately surrounding transcript. E-QGen [4] generates anticipated student inquiries from a lecture abstract via a synthesized lecture script, while Cardia et al. [3] used student personas to produce questions that instructors found difficult to distinguish from human-authored ones. This work moves AQG toward pedagogically plausible and student-like inquiry. However, the questions are still largely prepared outside the social flow of a live class. For QGen, relevance is not only a property of the generated text; it is also a situated relationship between the question, the unfolding lecture, and learners' in-the-moment uncertainty.

HCI research shifts attention from what AI generates to how learners and educators work with its output. DAPIE [22] turns children's why/how questions into stepwise explanatory dialogues, allowing learners to probe and refine an explanation rather than accept a single response. ReadingQuizMaker [30] and TutorCraftEase [20] similarly treat generated questions as editable candidates: AI initiates the process, but educators review, revise, and retain responsibility for the final result. This interactional view is increasingly expressed through feedback loops that connect conversation to evidence learners can inspect and act upon. CourseAssist [9] grounds tutoring in course materials and decomposes student requests to provide course-specific support. VisualCodeChat [24] integrates agentic exercises and dynamically generated algorithm visualizations, so conversational guidance is tied to representations that learners can examine. Wu et al. [43] connect contextual explanations and code suggestions to autograder feedback, while Li et al. [23] combine block programming, pedagogical dialogue, and embodied robot execution to make program behavior tangible while preserving learner control. These systems share an important principle: AI support becomes more educationally productive when learners can interrogate it, compare it with observable evidence, and decide how to revise their own work. QGen addresses an earlier and less structured point in this process. During a lecture, students may not yet have a code artifact, diagnosis, or even a well-formed question. The design challenge is therefore to make possible directions for inquiry visible without making the AI's formulation so complete that it substitutes for the learner's own metacognitive work.

That challenge is intensified in classrooms because asking a question is not only an individual sensemaking activity, but also a social act. Systems that adopt peer-like roles show how an agent's presence can shape participation. ClassMeta [29], for example, uses a virtual classmate to model participation and help break the ice in VR classrooms. Maiti and Goel [31] found that sustained interaction with the Jill Watson instructional partner corresponded with an increase in higher-order, Bloom-aligned student questions, suggesting that AI can influence learners' inquiry practices as well as provide information. Other work emphasizes the grounding and oversight required when AI enters an instructional setting. Inoue et al. [18] generate critical-thinking questions from lecture-slide structure and recommend relevant slides that learners can use to develop their inquiries. Miroyan et al. [34] evaluate responses to live student questions with teaching-assistant feedback, while Oversight in Action [36] requires an instructor to review course-grounded LLM drafts before they are posted. Ito et al. [19] brought question generation into the physical classroom through an LLM-driven robot whose timing and question types were controlled by the instructor. Although the robot supported engaging exchanges and reflection, students themselves still rarely asked questions, and some lower-level prompts were experienced as disruptive. These studies show that a plausible educational question can still be ineffective if it appears at the wrong time, assumes too much authority, or fails to reflect the group's needs. QGen therefore places AI-generated questions within a shared, instructor-facilitated process: their source remains visible, students can compare them with

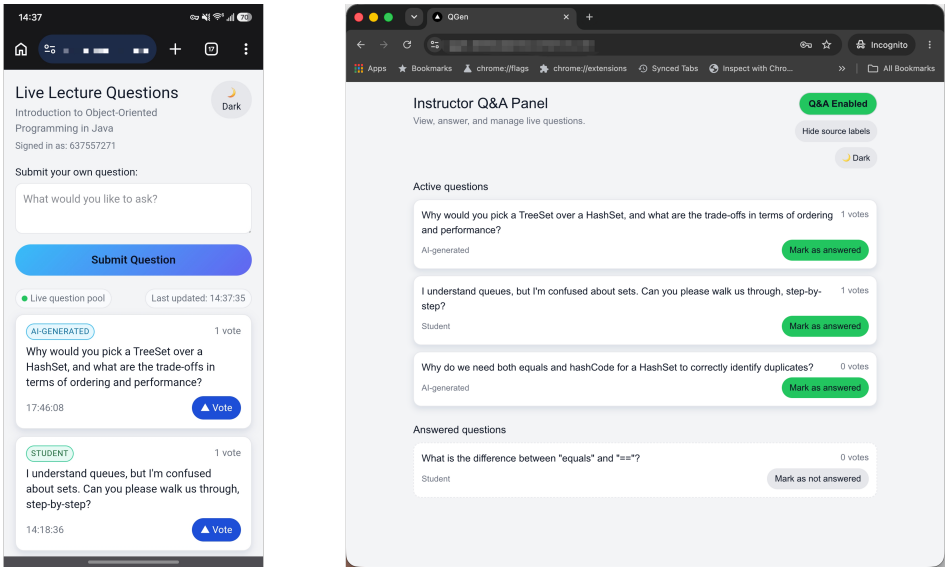


Figure 1: The student-facing (left) and instructor-facing (right) interfaces of the QGen system.

peer questions and vote on their relevance, and students retain a direct channel for expressing concerns the AI has not captured.

Taken together, prior work shows that live AI-supported questioning requires pedagogically plausible questions, interactions that preserve learner agency, and integration with classroom timing and social dynamics. Yet little research examines these requirements together by placing AI-generated questions alongside student-authored questions during a live lecture. QGen investigates this intersection.

3 QGen System Design

We designed QGen as a technology probe [17] to explore how AI-generated questions might shape students' question-asking practices during university lectures. Rather than embodying a finalized or optimized design, QGen was developed as a functional prototype to enable in-situ study of whether AI-generated questions (1) encourage students to ask their own questions, (2) help students reflect on what questions they might ask, and (3) prompt students to consider lecture material from alternative perspectives.

QGen generates questions dynamically during class lectures by recording and transcribing the instructor's speech in real-time and periodically prompting an LLM to generate questions using the cumulative lecture transcript. The system consists of three main components:

(1) An AI component that captures and continuously transcribes audio of the instructor's speech in real-time (captured from a wireless microphone that the instructor wears) and uses this continuously-updating transcript to periodically (i.e., once every two-and-a-half minutes) generate questions for the instructor from the perspective of a student.

(2) A student-facing mobile web interface (Figure 1, left) that allows students to view, vote on, and submit their own questions to the question pool in addition to the AI-generated questions.

(3) An instructor-facing web interface (Figure 1, right) that allows the instructor to review the questions in the pool (including student-submitted questions, AI-generated questions, and the number of votes each question has) and mark questions as answered whenever the instructor chooses to answer them.

The student-facing web interface is designed to be mobile friendly and easily accessed on students' personal devices (students can choose to access it on the browser of their smartphone, laptop, or tablet by scanning a QR code or entering a URL). A student logs in by entering their student ID number, which brings them to a screen where they can see the list of questions currently in the question pool. Each student can type in and submit their own questions and vote on as many questions as they want (including their own question(s), other students' questions, or AI-generated questions). Each student can vote on a single question at most once, and can also 'cancel' (i.e., retract) their vote on a question if they wish.

The instructor-facing web interface is similar to the student-facing interface, but it is intended to be used by the instructor on the lecture-podium computer, where they can display the list of questions to the students on the wall display (projector) at the front of the lecture room. The instructor can view the list of questions ordered from top-to-bottom based on the number of votes (highest-voted questions are displayed at the top) and mark questions as 'answered', where they are then removed from the question pool and moved to a different section of the interface (the instructor may also 'unanswer', i.e., unmark a question as answered, moving it back to the question pool and restoring its previous state).

On both the student- and instructor-facing interfaces, each question is presented with a label that indicates whether it was written

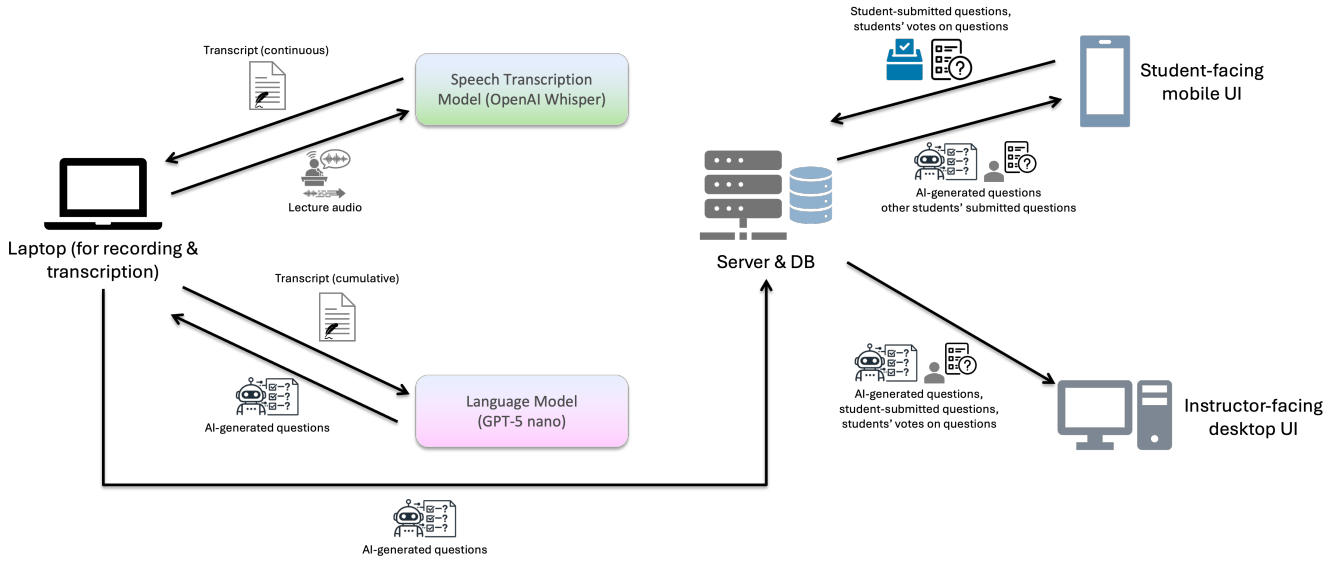


Figure 2: An overview of the QGen system architecture.

by a student or generated by AI. These labels do not indicate *which* student submitted a question; only whether a question was submitted by a student or generated by AI.

QGen was designed around the principle that AI-generated questions should scaffold, rather than replace, student inquiry. Its periodic AI-generated questions model possible lines of questioning that students can review, compare with their own understanding, and use as starting points for deeper inquiry. The student-facing interface preserves learner agency by allowing students to submit and vote on questions. The shared question pool also supports engagement: students can identify which questions peers value, instructors can address collectively prioritized needs, and AI-generated questions can lower the initial barrier to participation.

3.1 System Implementation

Figure 2 outlines the system implementation. The instructor- and student-facing web interfaces are hosted on Vercel and implemented in JavaScript using React.js. Questions and students' votes on questions are stored on a remote SQL database. Meanwhile, audio capture and transcription are handled by a laptop in the lecture room, and the instructor wears a wireless microphone that captures their voice and is wirelessly connected to the laptop. A Python script running on the laptop records the audio in overlapping 30-second segments (with 0.5-1 second of overlap each), filters these using voice-activity detection, then uploads these segments to OpenAI's Whisper [37] model (using OpenAI's API service) for transcription. Our internal testing of Whisper in our prototype revealed approximately 90–95% accuracy in transcribing pre-recorded lecture videos recorded by instructors in our department, and where transcription errors did occur, these were minor enough to not affect the overall quality of the questions generated by QGen. The text of the transcription result is then appended to the continuous lecture transcript, which is stored as a `.txt` file on the laptop. If the system is restarted mid-lecture (e.g., due to a crash or connectivity issue), it reloads the existing transcript and continues appending new content. At fixed intervals (once per two-and-a-half minutes), the laptop prompts GPT-5 nano (also using OpenAI's API service) with the full lecture transcript accumulated so far, along with a list of previously generated questions. The model is instructed to generate up to three new questions that either clarify potentially unclear content or invite reflection on the lecture material. See the supplementary materials for the full LLM prompt. Once the questions are generated, the laptop then writes these questions to the SQL database, and the student- and instructor-facing interfaces then display these questions.

4 Study Method: Deployment in Computer Science Lectures

Our goal was to explore and understand how AI can help students pose their own questions and reflect on learning material in classroom lectures. This study was approved by our university's research ethics review panel. Our research questions were as follows:

(1) How can an AI system that generates questions in live lectures affect students' learning experiences, including the types of questions that students ask and the ways they reflect on the learning material?
(2) What are students' perceptions of an AI system that generates questions in live lectures?
(3) What are students' interaction needs from an AI system that helps them construct questions and reflect on lecture material?

4.1 Setting and Participants

The deployment spanned four second-year undergraduate computer-science lectures, each two-hours long, at a university in China. The four lectures were delivered to two class groups (one with 254

students and another with 208 students) taking the same course (*"Introduction to Object-Oriented Programming in Java"*) and delivered across two weeks. Each class group learned the same content as the other during each week of the deployment.

In this course, before the QGen deployment, the students were normally given opportunities to ask questions both during and immediately after the lecture. Because of the class sizes, however, the students usually did not ask questions during the lecture, but on average about 5–10 students tended to ask questions immediately after each lecture by approaching the instructor once the lecture has ended.

In each lecture during the deployment, the instructor presented material for approximately 10–15 minutes until reaching a natural stopping point in the material. At this point, the instructor paused, displayed the instructor-facing interface with the list of questions on the projector, and gave students an opportunity to open the student-facing interface, read through the questions in the pool (including the AI-generated questions and those submitted by other students), submit questions of their own, and vote on questions.

After the submission and voting period, the instructor reviewed the ranked list and answered the two highest-voted questions that were relevant to the material just discussed. The instructor then marked addressed questions as answered, removing them from the student view while preserving them for the instructor's own reference. The lecture then continued with the next micro-lecture segment, repeating this cycle as time permitted.

During each lecture, a researcher on our team monitored the live transcription produced in real-time by QGen using the Whisper model to verify the quality of the transcription, and confirmed that no major transcription errors occurred during the deployment.

4.2 Data Collection and Analysis

At the end of the four-lecture deployment, students were invited to complete a survey about their experience with QGen, where they answered questions about their perceptions of AI-generated versus student-submitted questions and how the system shaped their own question asking and voting (see the supplementary materials for the full question list). A total of 215 students (each indicated as *'P#'* in the Findings) completed the survey. Students were allowed to answer the survey questions in English or Chinese.

We collected quantitative data to compare students' question voting patterns, preferences for voting for questions from other students vs. AI, and perceptions of questions from other students vs. AI reflecting one's learning needs. We compared students' vote counts on other students' questions vs. AI-generated questions using a negative binomial generalized linear mixed model with question source as a fixed effect and lecture session as a random intercept. This accounts for the count nature of the vote data and the clustering of questions within lecture sessions. Because the number of lecture sessions was small, we also inspected session-level mean differences as a sensitivity check.

We also collected qualitative open-ended responses on students' experiences and perceptions. We analysed the open-ended responses using thematic analysis [1] to characterize how AI-generated questions shaped students' reflection, question formulation, motivation, and perceived agency, as well as how students evaluated the fit of AI-generated questions to their in-the-moment learning needs.

Lastly, we coded the types of questions asked by students and generated by AI. The first two authors developed an initial codebook (Table 1), coded the questions into categories independently using this codebook, then iteratively discussed and refined their codes until they reached 100% agreement. We report on the categories of questions posed by students and AI as well as their relationships to students' learning needs in subsection 5.2.

5 Findings

5.1 Metacognition, Reflection of Learning Content, and Overreliance

Students reported positive opinions about the Q&A system and the AI-generated questions. Many students (e.g., P6, P17, P39, P96, P153) expressed that AI-generated questions helped them think about the learning content from different perspectives. For some, this helped them reflect on their own learning, including what they already knew about the course content and what knowledge they were still struggling to understand.

> *"AI will ask questions in combination with the content of the class, which usually allows me to think about what I have learned independently."* — P13

For others, this encouraged them to think more deeply about their own questions to ask.

> *"AI can generate questions from more angles and provide me with different perspectives to assist me in asking my own questions."* [Original: "AI能从更多角度生成问题，为我提供不同的思考角度，以此辅助我提出自己的问题。"] — P51

Some students were even motivated to ask their own questions simply by the mere presence of the AI-generated questions—e.g., expressing that they wanted to "compete" with the AI.

> *"AI-generated questions act as 'icebreakers'. When there is already a question on the screen, I feel that the mental barrier to submitting a question is lowered and I no longer worry about being the first to ask a question."* [Original: "AI 生成的问题起到了'破冰'的作用。当屏幕上已经有问题存在时，我会感觉提交问题的心理门槛降低了，不再担心成为第一个提问的人。"] — P24

While this was the case, some students felt that the AI-generated questions discouraged them from asking their own questions. For instance, the mere presence of AI-generated questions resulted in some students becoming less motivated to ask questions of their own, either because they feel they can 'relax' while the AI asks questions or they feel that the AI is already asking the types of questions they want to ask. Please see the attached supplementary materials for a list of the survey questions asked.

> *"My subconscious will think that AI will ask questions anyway, so can I relax a little?"* — P13

While some students expressed that the AI-generated questions helped them reflect more deeply on the material and come up with more of their own questions, others felt that the AI limited their

Question Category	Definition
Application	Questions about application, implementation, troubleshooting, etc.
Assessments	Questions about course assessments or requirements (e.g., exams, assignments, labs, etc.).
Clarification	Questions seeking clarification or more details on something already taught/covered.
Conceptual	Conceptual questions about underlying principles, relationships, mechanisms, or deeper meaning.
Metacognitive	Questions about learning strategies.
Speculation	Questions about hypothetical scenarios, consequences, or broader implications.
Unrelated	Questions completely unrelated to the learning material.

Table 1: The codebook of question types used to categorize questions posed by students and the AI.

own creativity in coming up with their own questions, as if it gave them tunnel vision.

> *"AI-generated questions can give you a clear thinking structure, but **they can also limit your own exploration**, making it either easier or harder to ask your own questions."* — P80

> *"Because AI may cover some more complex questions, which makes me to think about them rather than try to ask my own questions."* — P78

These responses reveal a limitation of AI scaffolding: when the scaffold appears too complete, students may treat it as a substitute for their own inquiry. This creates a tension between support and agency. AI-generated questions may be most productive when they support students' sense of competence while preserving autonomy over what to ask and how to ask it [39].

5.2 Question Types and Alignment with Learning Needs

Across the four lectures, students asked a total of 84 questions, while the AI posed a total of 57 questions (thus, the total number of questions across all four lectures was 141).

Students felt that the AI-generated questions were at times too advanced or focused on the wrong details of the lecture content.

> *"A few AI-generated questions were confusing and irrelevant because they focused on overly niche technical details that were not mentioned or emphasized in the lecture."* — P153

Many students expressed that they found student-submitted questions to be often more relevant than the ones generated by the AI, expressing that students' questions tended to better reflect the in-the-moment needs of students and consider the context of the entire course rather than just the single lecture.

> *"Questions submitted by students are more useful to me. Because classmates' questions often reflect real bottlenecks or logical contradictions we encounter in the actual understanding process, while AI questions are sometimes too perfect or textbookish and lack relevance."* [Original: "学生提交的问题对我更有用。因为同学的问题往往反映了我们在实际理解过程中遇到的真实瓶颈或逻辑矛盾，而 AI 的问题有时过于完美或教科书化，缺乏针对性。"] — P24

> *"The AI-generated questions are more focused on the classroom, and the students' questions are more focused on the actual situation."* [Original: "都有，AI生成的问题更聚焦课堂，同学的提问更聚焦实际情况。"] — P43

This was reflected by students' voting patterns. AI questions received on average (raw mean) 0.9 votes each (SD = 1.8), while student-submitted questions received on average (raw mean) 4.6 votes each (SD = 7.0). A negative binomial generalized linear mixed model with question source as a fixed effect and lecture session as a random intercept revealed a model-estimated mean vote count of 4.4 for student-submitted questions and 0.8 for AI-generated questions, and a significant effect of question source ($p < .001$).

Additionally, comparing the types of questions that students asked vs. those that the AI asked (see Figure 3), we found that students were more likely to ask questions related to assessments (upcoming exams and assignments), and occasionally questions unrelated to the course topic (e.g., playfully interacting with the QGen system). These questions together accounted for 69.9% of all questions asked by students, and 0% of the questions posed by AI. Given this, we were also interested in what types of questions students asked and voted for when *excluding* assessment-related and topic-unrelated questions—i.e., if students were not under pressure from course assessments and were strictly focused on learning, then what types of questions would they ask and vote for? When removing these questions (resulting in a new total of 26 questions asked by students and 57 questions asked by the AI; see Figure 4), we found that student-submitted and AI-generated questions appeared to more closely align, but students still appeared more likely to ask 'clarification' questions while the AI appeared more likely to ask 'application' questions. However, this could be more a reflection of the current configuration of QGen (e.g., how the LLM is prompted to generate questions).

We also redid the analysis of students' voting patterns after removing assessment-related and topic-unrelated questions. In this case, AI-generated questions received on average (raw mean) 0.9 votes each (SD = 1.8), while student-submitted questions received on average (raw mean) 2.7 votes each (SD = 2.9). After fitting a negative binomial generalized linear mixed model with question source as a fixed effect and lecture session as a random intercept, the model-estimated mean vote count was 2.1 for student-submitted questions and 0.9 for AI-generated questions. The effect of question source was still significant ($p = .011$).

These findings reveal that, with the current AI question-generation system, while AI-generated questions tended to act as good "icebreakers" and help some students reflect on the lecture material from different perspectives, students still tended to resonate more

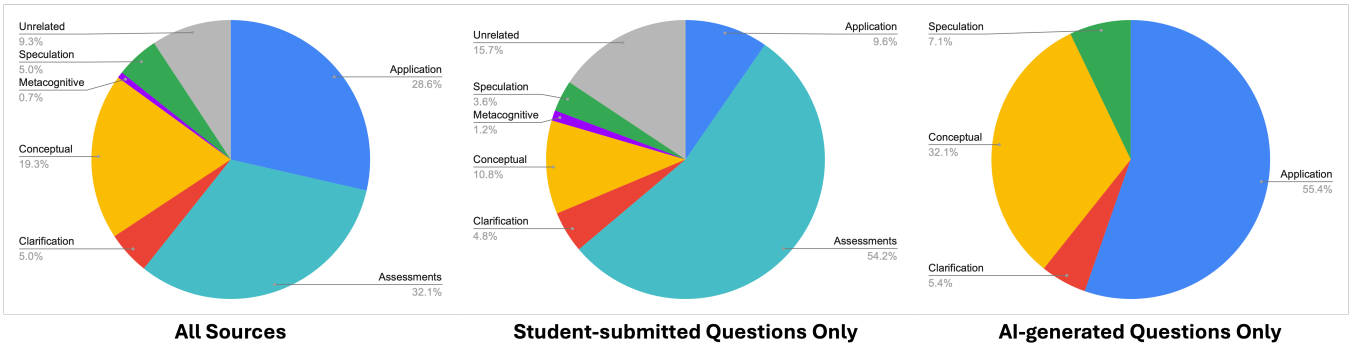


Figure 3: The distribution of question types from all sources (left), submitted by students (centre), and generated by AI (right).

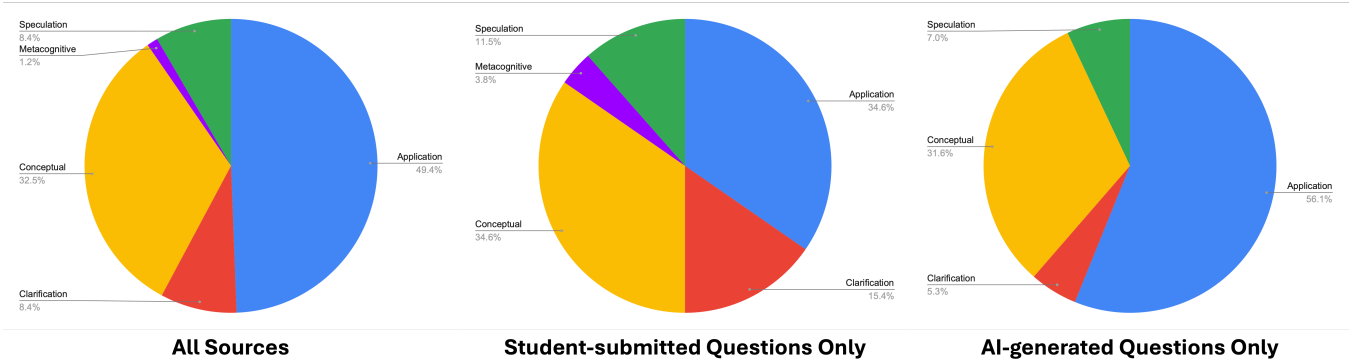


Figure 4: The distribution of question types from all sources (left), submitted by students (centre), and generated by AI (right); *excluding* questions related to assessments and questions unrelated to the course topic.

with the questions raised by other students. This suggests that, while the AI is capable of posing questions that are related to the learning material strictly from a *content* perspective, it still does not yet consider what students are currently struggling with or what their current learning needs or interests are.

5.3 Interactivity and Human-AI Collaboration

In its current state, the AI only listens to the instructor's speech and generates questions for the instructor periodically, without any further interaction with the students or the instructor. Some thought that this setup was too "rigid", and several participants expressed a desire to interact more with the AI beyond this—e.g., to influence the questions that the AI asks. Given the occasional misalignment between the questions the AI asks and the actual learning needs and interests of the students, some students wanted to influence the topics that the AI asked questions about, so that the AI would focus its questions on areas that are at the root of students' confusion.

> *"Let the AI adjust question topics based on previous user interactions (e.g., if a user struggled with a concept earlier, generate follow-up questions to reinforce it)."* — P147

Beyond just asking simple one-off questions, some students wanted the AI to act more akin to 'engaging in a conversation' with the instructor—for example, by asking follow-up questions based on the instructor's answers. Some students also wanted to be able to indicate to the AI which details or aspects to generate follow-up questions on.

> *"The system only generated questions based on the main lecture content, but it didn't allow me to request follow-up questions on a specific detail I cared about. This made it hard to dive deeper into personalized confusion."* — P147

6 Discussion and Design Recommendations

This study highlights both the potential benefits and challenges of deploying AI-generated questions in university lectures. While this study revealed that AI-generated questions could support deep reflection of the learning content for many students, it also revealed risks related to AI overreliance, misalignment with students' needs, and limited human-AI collaboration opportunities. In this section, we discuss design implications and recommendations for AQG systems that promote interactive collaboration with the AI and scaffold question asking, without undermining the student's own agency or reflection of the learning material.

6.1 Design AI as a Reflective Catalyst

Our findings suggest that AI-generated questions were valuable for prompting students to reflect on the lecture content from different perspectives, which some students suggested could help inspire them to formulate their own questions. However, some students reported reduced motivation to actively ask their own questions, as they perceived the AI as effectively asking questions for them.

In order to address this challenge of reduced motivation to ask one's own questions, AQG systems can **frame AI-generated questions as prompts for student reflection** on the learning material, rather than as replacements for students' own questions. Such systems can present AI-generated questions as *examples* or *starting points* for students' own questions, rather than as complete representations of what is worth asking. For example, AQG systems can visually frame AI-generated questions as 'possible directions' or 'starting prompts' rather than as complete questions. An AI system that encourages students to formulate their own questions can pair these initial question templates or candidates with lightweight prompts such as "What would *you* ask instead?" or "What do you think is missing from this question?"; or it could limit the perceived completeness of AI-generated questions (e.g., by generating intentionally partial or open-ended questions, for the user to fill in more details or decide on a more concrete direction). Such an approach could reframe the student's perception of the AI from being a replacement for students' questions to instead being a metacognitive scaffold, which could in turn address concerns around overreliance while amplifying the student's agency in asking the questions that align with their own needs.

6.2 Use Interaction Signals to Align Questions to Students' Needs, Not Just to Content

Our study revealed that students' own questions often reflected students' own needs and interests better than the AI-generated questions. In its current setup, QGen simply listens to the instructor's speech and responds by simply generating questions in response to the content of the instructor's speech. While this is a starting point, this simple one-turn interaction fails to consider the group dynamic of the class, nor does it promote a more collaborative or group-centred learning experience where the AI responds to the needs and signals from everyone in the class and the students contribute meaningfully to influencing the AI's questions.

This suggests that the AI should not necessarily try to compete with the student, but rather it should **make use of cues from the students** (such as students' questions, votes on questions, and other interaction signals) to **understand and assess students' needs**, and use this understanding to steer its question generation. For example, the AI could use early student-submitted questions to steer or constrain its own generated questions, generate explicit follow-up questions or prompts to students' questions or the instructor's answers, or analyse students' votes on questions to understand students' current learning needs and generate questions that are reflective of these needs. Additionally, students could be given the ability to 'mark' parts of the lecture that they are 'confused' by or want to enhance their understanding of, in order to signal to the AI to focus its question generation on these contents. Such techniques could not only make the AI more attentive to students' needs, but they could also result in an experience that more closely resembles a collaborative 'conversation' rather than a limited one-way response to the instructor's teaching that happens independently of students' experiences and contributions.

Interaction signals can also be understood as pedagogical diagnostics. Student questions, votes, confusion markers, and requests for follow-up can reveal what learners are currently able to understand with support. Instead of generating questions solely from instructor speech, future systems could use these signals to estimate where the class is struggling and then generate prompts that are closer to students' current learning needs.

6.3 Enable Students to Explicitly Steer AI Question Generation

The students in our study expressed a desire to influence the questions that the AI asked—e.g., by focusing on specific topics, adjusting question depth, or requesting follow-up questions on points of confusion. This suggests that, rather than treating question generation as a fixed background process, AQG systems should provide explicit mechanisms for students to steer the questions that the AI asks (in addition to the implicit steering through interaction signals suggested in subsection 6.2). Controls such as selecting question depth, allowing students to select the types of questions to ask (e.g., 'speculative', 'clarification', 'provocative', etc.), or requesting elaboration could serve as potential solutions to this.

7 Limitations and Future Work

While this work provides important insights, it is not without its limitations. For one, the deployment was conducted in a single institutional context and limited to four computer-science lectures. Student norms around participation, comfort with technology, and expectations of AI support may differ across disciplines, institutions, class sizes, and cultural settings. As such, findings around students' experiences, perceptions, and behaviours may not generalise directly to other educational contexts. Additionally, the instructor played an active role in pacing Q&A windows and selecting questions to answer. Different instructor facilitation styles or strategies could lead to different patterns of student engagement and interaction with the AI-generated questions. Lastly, the study focused on short-term classroom use and self-reported perceptions. While these data provide insight into students' experiences and behaviours during lectures, we did not measure longer-term learning outcomes, changes in question-asking skills, or shifts in student agency over time.

8 Conclusion

This work investigated how AI-generated questions impact students' experiences with question-asking in university lectures. Through the design, development, and deployment of QGen, we explored students' perceptions of AI-generated questions, their effects on students' own participation in Q&A, and their impacts on students' learning needs. Our findings show that AI-generated questions can support reflection and perspective-taking, and can lower barriers to participation. However, we also uncovered important tensions: some students experienced reduced motivation

to formulate their own questions, AI-generated questions were often misaligned with students' in-the-moment learning needs, and the current interaction model limited opportunities for meaningful human–AI collaboration. We therefore recommend, as future design opportunities, reframing the AI as a reflective scaffold rather than as a replacement for students' own question-asking, and leveraging interaction signals to better align AI-generated questions with learners' needs.

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Supplementary Materials – Survey Questions

A) Learning Experience and Understanding / 学习体验与理解

1. The system helped me better understand the lecture material.
   这个系统帮助我更好地理解了课堂内容。

2. Seeing AI-generated questions prompted me to think about the content from new or different perspectives.
   看到 AI 生成的问题促使我从新的或不同的角度思考课堂内容。

3. Seeing other students' questions prompted me to think about the content from new or different perspectives.
   看到其他同学提出的问题促使我从新的或不同的角度思考课堂内容。

4. The AI-generated questions highlighted aspects of the lecture I might have otherwise overlooked.
   AI 生成的问题指出了课堂中一些我可能会忽略的内容。

5. The other students' questions highlighted aspects of the lecture I might have otherwise overlooked.
   其他同学的问题指出了课堂中一些我可能会忽略的内容。

Open-Ended Questions (Optional) / 开放式问题（可选）

6. In what ways (if any) did AI-generated questions help you understand the lecture better?
   AI 生成的问题（如果有的话）是如何帮助你更好地理解课堂内容的？

B) Effects on Students' Own Question Asking / 对学生自主提问行为的影响

7. I felt confident submitting my own questions during the lecture.
   在课堂上，我对提交自己问题感到有信心。

8. The presence of AI-generated questions encouraged me to ask my own questions.
   AI 生成的问题鼓励我提出自己的问题。

9. The presence of other students' questions encouraged me to ask my own questions.
   其他同学的问题鼓励我提出自己的问题。

10. The presence of AI-generated questions discouraged me from asking my own questions.
    AI 生成的问题让我不太愿意提出自己的问题。

11. The presence of other students' questions discouraged me from asking my own questions.
    其他同学提出的问题让我不太愿意提出自己的问题。

12. I tended to vote for AI-generated questions.
    我倾向于为 AI 生成的问题投票。

13. I tended to vote for other students' questions.
    我倾向于为其他同学提出的问题投票。

14. I felt that the AI-generated questions were similar to the types of questions I would ask.
    我觉得 AI 生成的问题与我会提出的问题类型相似。

15. I felt that the other students' questions were similar to the types of questions I would ask.
    我觉得其他同学提出的问题与我会提出的问题类型相似。

16. I felt that the AI-generated questions were too advanced compared to my own thinking.
    我觉得 AI 生成的问题相比我的思考水平来说过于高级。

17. I felt that the AI-generated questions were too basic compared to my own thinking.
    我觉得 AI 生成的问题相比我的思考水平来说过于简单。

18. The AI-generated questions made me feel that my own questions were unnecessary or less valuable.
    AI 生成的问题让我觉得我自己的问题没有必要或没有价值。

19. The other students' questions made me feel that my own questions were unnecessary or less valuable.
    其他同学提出的问题让我觉得我自己的问题没有必要或没有价值。

Open-Ended Questions (Optional) / 开放式问题（可选）

20. Explain why AI-generated questions made it easier or harder for you to ask your own questions.
    请解释为什么 AI 生成的问题让你更容易或更难提出自己的问题。

21. How did the AI-generated questions influence the kinds of questions you thought about asking?
    AI 生成的问题如何影响你思考要提出哪类问题？

C) Interaction with the Instructor / 与教师互动

22. The question system made the instructor more understanding of the students' needs.
    这个提问系统让教师更了解学生的需求。

23. The AI-generated questions made the instructor more understanding of the students' needs.
    AI 生成的问题让教师更了解学生的需求。

D) Comparative Questions / 比较性问题

24. Compared to a system with only student-submitted questions, mixing AI-generated and student-submitted questions helped me reflect more deeply on the lecture content.
    与只包含学生提交问题的系统相比，混合 AI 生成的问题和学生提交的问题帮助我更深入地反思课堂内容。

25. Compared to a system with only student-submitted questions, mixing AI-generated and student-submitted questions made me more likely to participate (view, vote on, and submit questions).
    与只包含学生提交问题的系统相比，混合 AI 生成的问题和学生提交的问题让我更愿意参与（浏览、投票和提交问题）。

Open-Ended Questions (Optional) / 开放式问题（可选）

26. What types of questions (AI-generated vs. student-submitted) felt more useful to you during learning, and why?
    在学习过程中，你觉得哪类问题（AI 生成 vs. 学生提交）对你更有帮助？为什么？

E) Perceived Quality and Relevance of AI Questions / 对 AI 问题质量和相关性的看法

27. The AI-generated questions felt relevant to the content being taught.
    AI 生成的问题与课堂内容相关。

28. Some AI-generated questions felt off-topic or misaligned.
    有些 AI 生成的问题感觉不太相关或偏离主题。

29. The AI-generated questions helped surface important ideas I had not considered.
    AI 生成的问题帮助我发现了一些我之前没有考虑过的重要内容。

Open-Ended Questions (Optional) / 开放式问题（可选）

30. Which AI-generated question(s) did you find most helpful, and why?
    你觉得哪些 AI 生成的问题最有帮助？为什么？

31. Were any AI-generated questions confusing, unhelpful, or irrelevant? Please explain.
    是否有 AI 生成的问题让你感到困惑、不太有用或不相关？请说明。

F) Future Improvements and User Experience Feedback / 未来改进与用户体验反馈

Open-Ended Questions (Optional) / 开放式问题（可选）

32. What features would you most like to see added to this system?
    你最希望这个系统增加哪些功能？

33. What would make you more likely to participate in question-asking during lectures?
    什么能让你在课堂上更愿意参与提问？

34. What frustrated you or felt missing while using this system?
    在使用这个系统的过程中，有什么让你感到困惑或觉得缺失的部分？

35. What advice would you give to improve the AI-generated questions?
    你对改进 AI 生成的问题有什么建议？

36. Do you have any additional thoughts you would like to share?
    你还有其他想分享的想法吗？

Supplementary Materials – QGen Question Generation LLM Prompt

```
You are an undergraduate student taking a course on
{course_title_or_topic}. You are in class observing and focusing on a
lecture being given by the professor. Your task is to generate
questions to ask the professor in class, based on the lecture content.

The generated questions should each be at most one sentence long, and
should be simple (i.e., ask questions that a first- or second-year
undergraduate student would ask). Avoid overly complex or technical
questions (e.g., avoid questions that go far beyond the course
material). Imagine that you are an student who is new to the course
topic, and you are trying to understand the lecture content better.
Phrase your questions as though you are a young undergraduate student
(i.e., avoid super formal language -- use casual and straightforward
wording).

The following is the transcript of lecture so far:

```
{lecture_transcript}
```

The following is a list of the questions previously generated from
this lecture -- generate new questions rather than duplicates. If you
wish, you may generate questions that act as follow-up questions to
previous questions:

```
{prev_q_text}
```

Generate three (3) new questions adhering to the above requirements.
Make sure the questions are relevant to the lecture content provided.

Output the generated questions as a JSON adhering to the following
schema (include *only* the new questions in the JSON output):
{{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "QuestionsList",
  "type": "object",
```

```
  "properties": {{
    "questions": {{
      "type": "array",
      "description": "A list of questions generated based on the
lecture content.",
      "items": {{
        "type": "string"
      }}
    }}
  }},
  "required": ["questions"],
  "additionalProperties": false
}}
```